Exp: 4 Design a lexical Analyzer to validate operators to recognize the operators +,-,*,/ using regular arithmetic operators using C

```
#include <stdio.h>

#include <string.h>

int main() {

   char s[5];

   printf("Enter any operator: ");

   if (fgets(s, sizeof(s), stdin) != NULL) {

      s[strcspn(s, "\n")] = '\0';


      switch(s[0]) {

         case '>':

            if(s[1] == '=') printf("Greater than or equal\n");

            else printf("Greater than\n");

            break;

         case '<':

            if(s[1] == '=') printf("Less than or equal\n");

            else printf("Less than\n");

            break;

         case '=':

            if(s[1] == '=') printf("Equal to\n");

            else printf("Assignment\n");

            break;

         case '!':

            if(s[1] == '=') printf("Not Equal\n");

            else printf("Bit Not\n");

            break;

         case '&':

            if(s[1] == '&') printf("Logical AND\n");

            else printf("Bitwise AND\n");

            break;
```

```
            case '|':
                if(s[1] == '|') printf("Logical OR\n");
                else printf("Bitwise OR\n");
                break;
            case '+': printf("Addition\n"); break;
            case '-': printf("Subtraction\n"); break;
            case '*': printf("Multiplication\n"); break;
            case '/': printf("Division\n"); break;
            case '%': printf("Modulus\n"); break;
            default:  printf("Not an operator\n");
        }
    }
    return 0;
}
```

Output:

```
Enter any operator: >=
Greater than or equal


...Program finished with exit code 0
Press ENTER to exit console.
```